Chapter
4

Microbial Metabolism

Learning Objectives

After studying this chapter the students will be able to,

- *Identify the role of ATP in cellular activities.*
- *Define metabolism and describe the fundamental differences between catabolism and anabolism.*
- *Explain oxidation – reduction reaction.*
- *List and provide examples of three types of phosphorylation reactions that generates ATP.*
- *Describe the Carbohydrate, Lipid, Protein and its pathways (Glycolysis, Krebs cycle, electron transport chain)*
- *Electron transport chain and chemiosmotic model for ATP generation.*
- *Understand about the types of fermentation and its products.*
- *Describe the mechanism of enzymatic activity and significance of microbial enzymes.*

Chapter Outline

4.1 Metabolism

4.2 Energy of Chemical Reaction

4.3 Generation of ATP

4.4 Carbohydrate Catabolism

4.5 Tricarboxylic Acid Cycle

4.6 Electron Transport Chain

4.7 Lipid Catabolism

4.8 Protein Metabolism

4.9 Fermentation

4.10 Enzymes

KJ1U4L

All living organisms are constantly in need of energy to function. The life support activity of even the most structurally simple organism involves a large number of complex biochemical reactions. Living cells carry out three major types of processes namely **Chemical Process, Transport Process** and **Mechanical Process**. In chemical processes, energy is required to synthesize complex biological molecules from much simpler molecules. Transport processes require energy to take up nutrients, eliminate waste, and maintain ion balance. Mechanical processes require energy to change the physical location of structures within cells. Even during resting state, a substantial amount of energy is needed for fundamental functions of cells. All living system obeys the laws of thermodynamics. This law analyzes energy

TN_GOVT_XII_Micro_Biology_CH04.indd 45 04-03-2019 12:54:52